



MEHMET TURAN YARDIMCI

Date of birth: 01/11/2003 | **Nationality:** Turkish | **Gender:** Male | **Address:** Türkiye (Home)

EDUCATION & TRAINING

01/10/2021 - 03/10/2025 - ADANA, TÜRKİYE

BACHELOR'S DEGREE IN COMPUTER ENGINEERING (ENGLISH)- ÇUKUROVA UNIVERSITY

- Academic Recognition: Honor Student (Fall 2023-2024), High Honor Student (Fall 2024-2025)
- Programming and algorithms (object-oriented programming, data structures, algorithms, programming languages, compiler construction)
- Computer systems and hardware (digital design, microprocessors, computer organization, circuit theory)
- Artificial intelligence and machine learning (AI systems, pattern recognition, optimal control, reinforcement learning)
- Networks and communication (computer networks, internet communication, operating systems)
- Mathematics and theory (linear algebra, differential equations, probability and statistics, discrete mathematics, theory of computation)
- Robotics and human-computer interaction (informatics in robotics, HCI, computer graphics)
- Software engineering (requirements analysis, software architecture, design patterns, database management)

Level in EQF: 6

27/09/2024 - 11/02/2025 - BIAŁYSTOK, POLAND

ERASMUS+ EXCHANGE PROGRAM IN COMPUTER SCIENCE- BIALYSTOK UNIVERSITY OF TECHNOLOGY

- Robotics and automation (PID control, sensor integration, robot programming, control systems)
- Human-computer interaction (VR/AR techniques, usability engineering, user interface design)
- Software development (object-oriented programming in Java, business applications, mobile systems)
- Computer graphics (2D/3D rendering, geometric transformations, color models, visualization)

Level in EQF: 6

WORK EXPERIENCE

18/08/2025 - 12/09/2025 - ADANA, TÜRKİYE

COMPUTER ENGINEERING INTERNKIVANÇ TEKSTİL

- Supported in-house software development processes using .NET and C#.

14/07/2025 - 11/08/2025 - SILIFKE, TÜRKİYE

COMPUTER ENGINEERING INTERNMEDCEM

- Supported in-house software development using .NET and C#.
- Designed and implemented algorithmic solutions.
- Gained hands-on experience in software architecture, debugging, and UI design.

PUBLICATIONS

Critic Architecture Matters: Dual vs. Unified Critics for Humanoid Loco-Manipulation 2026

Accepted as poster presentation at IEEE International Conference on Robotics and Automation (ICRA) 2026 Workshop, Vienna, Austria, June 1-3, 2026.

Yardımcı, M.T. ICRA 2026 Workshop on Reinforcement Learning in the Era of Imitation Learning (RL4IL)

Links

https://www.linkedin.com/posts/mehmetturanyardimci_robotics-reinforcementlearning-isaac-lab-activity-7463248002841944065-9C83?utm_source=share&utm_medium=member_desktop&rcm=ACoAADg5-uMBdrgFmv4IXpG3VqvkpvUmzqmnyvM | <https://github.com/mturan33/isaac-g1-ulc> | <https://openreview.net/forum?id=QTQk2HICbs> | <https://rl4il-icra.github.io/>

Benchmarking Local Path Planners in ROS using the BARN Dataset 2026

Pending publication. Comparative benchmark of TEB, DWA, MPC, and Lattice planners on the BARN navigation dataset.

Yardımcı, M.T., Çoğurcu, Y.E. Çukurova University Journal of the Faculty of Engineering

Link <https://github.com/mturan33/benchmark-local-path-planners-barn-challenge>

PROJECTS

01/04/2026 - CURRENT

G1 Vision-Language-Action (VLA) Pipeline - Building RL-to-IL-to-VLA pipeline for Unitree G1 humanoid robot.

- Collecting expert demonstrations from trained PPO policies, distilling into end-to-end visuomotor policies.
- Pipeline: RL expert rollout LeRobot v2 dataset ACT/Diffusion Policy GR00T N1.6 fine-tuning.
- Joint-level 22-dim action space (15 locomotion + 7 arm), RGB observations from tiled camera.
- Goal: End-to-end vision-to-action policy without hand-crafted skill primitives.
- Technologies: NVIDIA Isaac Lab, LeRobot, ACT, Diffusion Policy, GR00T N1.6, PyTorch

Link <https://github.com/mturan33/isaac-g1-vla>

01/02/2026 - 30/04/2026

G1 Humanoid Hierarchical Control Pipeline (HCP) - End-to-end hierarchical control system for Unitree G1 humanoid: VLM task planning + RL locomotion + arm manipulation + drawer interaction.

- High-level: VLM planner (Qwen3-VL) generates skill sequences from natural language instructions.
- Low-level: PPO-trained locomotion (17k steps/sec) + arm policy (3cm precision).
- Skills: Walk, reach, grasp, pull drawer, pick-and-place. Achieved 6/6 drawer success, 8/8 pick-and-place success.
- Technologies: Isaac Lab, RSL-RL, PyTorch, Ollama, Qwen3-VL, CUDA

Link <https://github.com/mturan33/isaac-g1-hierarchical>

01/12/2025 - 28/02/2026

Unitree G1 Locomotion Control (ULC) — Multi-stage RL Training - Multi-stage PPO training pipeline for Unitree G1 whole-body locomotion.

- 5-stage curriculum: Flat walking Velocity tracking Terrain adaptation Torso stabilization Arm coordination.
- Achieved 17,000+ steps/sec on RTX 5070 Ti with 4096 parallel environments, L7 curriculum.
- Locomotion policy: 99.6% success rate.
- Technologies: NVIDIA Isaac Lab, RSL-RL, PyTorch, CUDA

Link <https://github.com/mturan33/isaac-g1-ulc>

01/11/2025 - 01/12/2025

Isaac Lab Anymal-C Quadruped Locomotion Training - Implemented PPO algorithm from scratch for Anymal-C quadruped robot in NVIDIA Isaac Lab environment.

- Achieved 17,000+ steps/second training speed on RTX 5070 Ti GPU.
- Professional-grade performance with domain randomization, reward shaping, and vectorized environments.
- Technologies: Isaac Lab, RSL-RL, PyTorch, CUDA, TensorBoard

Link <https://github.com/mturan33/isaacclab-anymal-locomotion>

01/09/2025 - 01/10/2025

MuJoCo Ant-v5 PPO Implementation - Implemented PPO and SAC algorithms from scratch using pure NumPy and PyTorch for MuJoCo Ant-v5 environment.

- Achieved 2700+ reward by eliminating hopping behavior through custom reward shaping.
- Integrated parallel environments (16 env), GAE, observation normalization, and learning rate annealing.
- Technologies: Python, PyTorch, NumPy, MuJoCo, TensorBoard

Link <https://github.com/mturan33/mujoco-ant-ppo>

01/08/2025 - 01/09/2025

Live Actor-Critic Training for CartPole - Created an interactive Streamlit web application demonstrating real-time Reinforcement Learning training.

- Implemented Actor-Critic algorithm with adjustable hyperparameters for educational purposes.
- Technologies: Python, PyTorch, Streamlit, Gymnasium

Link <https://github.com/mturan33/my-actor-critic>

01/11/2024 - 01/01/2025

PID Control Algorithm Implementation with NXT Robot - Implemented PID control algorithm with real-time sensor feedback on LEGO

Mindstorms NXT robots.

- Technologies: NXC, Bricx Command Center, PID Controller

Link https://github.com/mturan33/PID_Implementation_With_NXT_Robot

01/11/2023 - 01/06/2024

YOLOv7 Fixed-Wing UAV Detection System - Designed and trained YOLOv7-based real-time object detection system for fixed-wing UAV applications.

- Integrated with Jetson Nano + Pixhawk hardware for autonomous navigation tasks.

- Part of TEKNOFEST autonomous UAV competition with AGM Alkar UAV Team.

Link https://github.com/mturan33/YOLOV7-Fixed-Wing_UAV_Detection

VOLUNTEERING

01/02/2022 - 01/10/2024 Adana, Türkiye

1.5 Adana AGM Alkar UAV Team - Led a team of 10+ members designing autonomous UAV systems.

- Integrated YOLOv5, Jetson Nano, and Pixhawk for TEKNOFEST competitions.

- Designed AI-based image processing modules for autonomous navigation.

LANGUAGE SKILLS

Mother tongue(s): **TURKISH**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

CERTIFICATIONS

24/02/2024 Coursera

Supervised Machine Learning: Regression and Classification

Online -

Link <https://coursera.org/share/c5a7b4bf1772c264f618f507c41940c4>

SKILLS

Robot Learning

Reinforcement Learning | Reward Shaping | Domain Randomization | PPO | Actor-Critic | Curriculum Learning | SAC | GAE | Imitation Learning

Vision-Language-Action Models

SmolVLA | LeRobot Framework | OpenVLA | GR00T N1.6 | Flow Matching | ACT | Diffusion Policy

Hierarchical Control

VLM Planning | Whole-Body Loco-Manipulation | Skill Libraries | Closed-Loop Replanning | Qwen3-VL | Florence-2

Programming

Python | C/C++ | .NET / C# | NumPy | Pandas | CUDA | Git | Linux | Ubuntu | Conda

Simulation

NVIDIA Isaac Lab | NVIDIA Isaac Sim | MuJoCo | Gazebo | RViz | ROS | ROS2

AI/ML

PyTorch | TensorBoard | YOLO | OpenCV | CNNs | Transformers | Hugging Face | Matplotlib

Robotics

PID Control | Jetson Nano | Pixhawk | Unitree G1 Humanoid | Universal Robots UR5e | Robotiq Gripper